

SPIRITWOOD WEST, SK, Canada

WMO: 711330

Lat: 53.3661N

Lon: 107.5564W

Elev: 1917

StdP: 13.71

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.1	-24.0	-36.6	0.7	-29.8	-30.1	1.1	-23.7	25.6	20.1	22.6	15.8	4.1	160	0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	82.2	65.1	78.9	63.7	76.0	62.2	69.1	77.6	66.6	75.1	64.4	72.8	10.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	103.1	74.1	63.5	94.2	70.8	61.3	86.9	68.6	34.5	77.7	32.3	75.2	30.6	72.7	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	19.8	17.7	DB	-37.9	88.6	5.9	4.5	-42.1	91.8	-45.6	94.4	-48.9	97.0	-53.2	100.2	(4)
			WB	-37.9	73.0	5.7	3.2	-42.0	75.2	-45.3	77.1	-48.5	78.9	-52.7	81.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	4.8	8.0	19.4	36.4	49.1	57.6	62.0	59.6	50.9	37.6	20.9	8.2	(6)
(7)		DBStd	23.43	16.67	14.38	14.42	10.44	8.26	5.98	5.07	5.86	7.77	8.87	12.45	14.97	(7)
(8)		HDD50	6719	1403	1176	947	419	117	9	0	4	82	393	872	1297	(8)
(9)		HDD65	11119	1868	1596	1412	858	495	230	119	183	426	848	1322	1762	(9)
(10)		CDD50	1130	0	0	0	11	90	237	372	303	107	10	0	0	(10)
(11)		CDD65	56	0	0	0	0	2	9	26	17	2	0	0	0	(11)
(12)		CDH74	1093	0	0	0	5	79	173	395	335	103	3	0	0	(12)
(13)		CDH80	207	0	0	0	1	15	32	67	73	19	0	0	0	(13)
(14)	Wind	WSAvg	8.2	8.1	8.0	8.5	9.1	9.2	8.2	7.5	7.1	8.0	8.6	8.3	7.9	(14)
(15)	Precipitation	PrecAvg	18.0	0.8	0.6	0.7	1.2	1.6	3.3	3.4	2.4	1.6	1.0	0.8	0.7	(15)
(16)		PrecMax	24.8	1.4	1.2	1.3	3.2	4.1	5.5	5.5	4.7	3.7	2.5	1.9	1.5	(16)
(17)		PrecMin	12.6	0.3	0.2	0.1	0.2	0.3	1.5	1.5	0.5	0.4	0.2	0.2	0.3	(17)
(18)		PrecStd	2.6	0.3	0.2	0.3	0.7	0.9	1.1	1.0	0.9	0.5	0.4	0.4	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	41.6	40.7	51.3	71.1	81.7	83.7	86.1	87.4	82.8	70.7	52.1	42.7	(19)	
(20)		MCWB	36.2	35.6	42.6	51.9	58.5	64.2	69.1	67.2	62.5	55.4	44.0	36.6	(20)	
(21)		DB	37.9	36.4	45.0	64.5	75.2	78.5	81.4	81.5	76.7	62.1	45.5	38.0	(21)	
(22)		MCWB	34.1	32.4	38.5	48.6	55.0	62.0	67.5	65.4	59.3	50.1	39.5	34.1	(22)	
(23)		DB	33.8	32.6	41.4	59.0	71.0	74.9	78.4	77.6	71.5	56.6	40.7	33.8	(23)	
(24)		MCWB	31.0	29.8	36.3	46.1	53.4	60.2	65.7	63.9	57.0	46.6	36.2	30.8	(24)	
(25)		DB	29.4	29.2	38.0	53.4	66.8	71.7	75.4	74.1	66.4	51.8	36.6	29.8	(25)	
(26)		MCWB	27.1	27.2	34.1	43.6	51.8	58.7	64.1	62.4	54.7	44.8	33.6	27.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	37.0	36.2	43.2	53.6	61.5	69.0	73.4	70.7	65.2	56.3	44.3	37.6	(27)	
(28)		MCDWB	40.9	40.1	50.3	68.8	73.0	78.6	81.1	80.3	75.5	68.5	51.6	41.2	(28)	
(29)		WB	34.4	32.7	39.1	50.3	57.6	65.2	70.3	67.8	61.1	51.3	40.4	34.3	(29)	
(30)		MCDWB	37.6	35.8	44.3	62.0	70.7	73.9	78.0	77.4	72.8	59.2	44.9	37.7	(30)	
(31)		WB	31.1	30.0	36.7	47.1	55.3	62.8	67.7	65.5	58.2	47.8	36.8	30.9	(31)	
(32)		MCDWB	33.8	32.4	40.7	57.0	67.8	70.8	75.5	74.8	69.0	55.5	40.2	33.8	(32)	
(33)		WB	27.1	27.1	34.3	44.0	52.8	60.4	65.5	63.3	55.7	44.9	33.8	27.5	(33)	
(34)		MCDWB	29.3	29.3	37.7	52.6	64.4	68.7	73.1	72.2	64.8	50.7	36.1	29.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	18.2	20.1	19.5	20.2	23.9	21.3	22.1	23.3	22.8	18.7	15.5	17.2	(35)	
(36)		5% DB	19.1	21.6	19.7	29.2	31.5	28.3	27.9	30.4	32.9	28.3	20.2	19.6	(36)	
(37)		MCWBR	16.3	18.2	14.4	16.4	15.6	14.4	15.9	16.4	17.6	17.2	14.8	16.5	(37)	
(38)		5% WB	19.0	21.2	19.0	27.4	27.6	23.8	25.5	27.2	29.6	26.4	19.1	18.9	(38)	
(39)		MCWBR	16.5	18.2	14.3	16.4	15.0	14.2	16.5	16.3	17.1	16.8	14.7	16.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.233	0.256	0.277	0.327	0.345	0.363	0.363	0.359	0.308	0.276	0.244	0.224	(40)	
(41)		taud	2.333	2.305	2.369	2.390	2.388	2.371	2.376	2.378	2.526	2.578	2.466	2.345	(41)	
(42)		Ebn at Noon	247	274	292	288	286	281	279	273	278	268	245	230	(42)	
(43)		Edh at Noon	17	25	30	33	35	36	36	33	25	19	15	14	(43)	
(44)	All-Sky Solar Radiation	RadAvg	308	646	1137	1512	1822	1837	1888	1556	1092	633	287	218	(44)	
(45)		RadStd	29	35	76	141	154	144	123	128	124	104	43	25	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page